# CAREER AUTHENTICITY ENGINE

A Comprehensive Deep-Dive into Multi-Layered Job Fraud Detection

**Theoretical Framework & Technical Implementation Guide**

Prepared for Technical Review & Academic Defense

# 1. Executive Summary

The job market today faces a critical crisis of trust. As recruitment moves entirely online, the barrier to entry for scammers has dropped. The **Career Authenticity Engine** addresses this by shifting the paradigm from simple human observation to **Automated Multi-Source Intelligence**.

## Core Problem Statement

Scammers are now using Generative AI to write "perfect" job descriptions. Consequently, keyword-based detection is obsolete. Our system solves this by analyzing the "Digital Fingerprint" of the employer across legal, social, and structural domains.

## The 4 Pillars of Authenticity

- **Linguistic Integrity:** Does the text structure match known fraud patterns?
- **Legal Accountability:** Does the company exist in Government (MCA) registries?
- **Infrastructure Maturity:** Is the website a throwaway domain or a long-standing asset?
- **Social Proof:** Is there a verifiable human network (LinkedIn) behind the brand?

# 2. Detailed Technical Pipeline

The workflow is designed as a linear chain of trust, where each link adds more data to the final calculation.

### PHASE 1: REAL-TIME DOM INTERCEPTION

The `content.js` script acts as a sensor. It utilizes MutationObservers to wait for the job posting to load on the screen. It then performs "Semantic Peeling" – removing site-specific artifacts (like Google Jobs headers or Indeed sidebars) to isolate the raw employer identity.

> **The "Skima.ai" Challenge:** When scanning a platform URL like `nykaa.skima.ai`, the engine performs "Domain Peeling" to recognize that *Nykaa* is the subject, not the platform provider.

### PHASE 2: THE ANALYSIS HUB (FLASK)

The backend receives the JSON payload and spawns multiple asynchronous threads to minimize user wait time. One thread handles the **ANN (Deep Learning)**, while others ping external APIs for **MCA Master Data** and **WHOIS info**.

# 3. Deep Learning Architecture (ANN)

The "Brain" of our system is an Artificial Neural Network (specifically a Multi-Layer Perceptron).

### FEATURE ENGINEERING: TF-IDF

We use TF-IDF (Term Frequency-Inverse Document Frequency) to vectorize job text. Unlike simple word counting, TF-IDF reduces the weight of common words ("the", "and") and amplifies "signature" words used by scammers.

$$TF\text{-}IDF(t, d) = TF(t, d) \times log(N / DF(t))$$

### ANN LAYER DESIGN

- **Input Layer (5000+ Neurons):** Receives the TF-IDF vector.
- **Hidden Layer 1 (128 Neurons - ReLU):** Identifies primary patterns (e.g., specific salary claims).
- **Hidden Layer 2 (64 Neurons - ReLU):** Identifies correlations (e.g., "Urgent" + "WhatsApp" + "Deposit").
- **Output Layer (1 Neuron - Sigmoid):** Compresses signals into a 0.0 to 1.0 Scam Probability.

$$Output = \sigma(\Sigma\, w_i x_i + b)$$

# 4. Verification Layers & APIs

This is where the "Theory" meets "Hard Evidence."

## Government Registry (The Sandbox API)

Every legal business in India must have a CIN. The engine cleans the company name and queries the MCA database. This is a **binary trust anchor**: if a company claims to be a Private Limited entity but has no CIN, the trust score is penalized by 40% immediately.

## Infrastructure Audit (WHOIS & SSL)

We analyze the age of the domain. A domain registered less than 30 days ago is a "high-entropy" signal. Coupled with a "Free" SSL certificate (e.g., Let's Encrypt), this signals a temporary scam portal.

## Global Blacklisting (MongoDB Atlas)

Every scan result is hashed and stored in the cloud. If multiple users scan the same suspicious domain, the **Global Network Effect** kicks in, and the site is auto-blacklisted for all future users.

# 5. Possible Examination Questions

*Comprehensive study of potential questions a teacher or evaluator might ask.*

### Q1: Why use an ANN instead of a simple Keyword Filter?

A: Scammers use 'Cloaking' (replacing flagged words with synonyms). ANN identifies the **structural context** and hidden relationships between words that a keyword list cannot see.

### Q2: How do you handle 'False Positives' for new startups?

A: We use Weighted Math. A low domain age (Infra) is offset by a valid MCA Registration (Legal). A startup will have a CIN, while a scammer will not.

### Q3: What is the significance of the 'Sigmoid' function in your output layer?

A: Sigmoid squashes the neural signals into a range between 0 and 1, which perfectly represents a 'Probability' score.

### Q4: Explain the importance of 'CORS' in this project.

A: CORS (Cross-Origin Resource Sharing) allows the browser extension to securely send data to a different domain (the local Flask server) without being blocked by browser security.

# 6. Deep-Tech Q&A

### Q5: Why is 'Certifi' used in your MongoDB connection?

A: Python on Windows often lacks updated root certificates. Certifi provides the verified SSL certificates required to establish a secure TLS handshake with MongoDB Atlas.

### Q6: What happens if an API (like Sandbox) is down?

A: The code uses 'Try-Except' blocks to ensure **Graceful Degradation**. The system will skip that layer and re-calculate the score based on remaining data instead of crashing.

### Q7: How did you generate the dataset for training the ANN?

A: We utilized 'Synthetic Data Generation'—taking thousands of legitimate job posts and injecting 'Scam noise' (e.g., Telegram links, urgent deadlines) to train the model on negative samples.

# 7. Limitations & Drawbacks

No system is perfect. Acknowledge these during a defense to show technical maturity.

- **API Latency:** Relying on 4-5 external APIs can slow down the scan time to 3-5 seconds.
- **Dynamic IP Issues:** MongoDB Network Access requires the user's IP to be whitelisted, which is difficult for mobile users.
- **Language Barrier:** The current ANN is trained on English text and may struggle with jobs written in regional languages.
- **Subdomain Camouflage:** Scammers using high-authority domains (like `sites.google.com/scam`) can trick the Domain Age check because they "borrow" Google's age.

# 8. Future Roadmap

How we can take this project from a prototype to a market-ready security product.

### 1. BI-LSTM IMPLEMENTATION

Upgrade the ANN to a Recurrent Neural Network (RNN) that understands the sequence of words even better than TF-IDF.

### 2. IMAGE-BASED VERIFICATION

Use OCR (Optical Character Recognition) to scan job posters/images for hidden WhatsApp numbers often missed by text scrapers.

### 3. BLOCKCHAIN VERIFICATION

Store company credentials on a decentralized ledger so that their identity cannot be spoofed or altered.

### 4. MOBILE APP INTEGRATION

Build a mobile version that scans job messages on WhatsApp and Telegram directly.

# 9. Final Conclusion

The **Career Authenticity Engine** is more than just code; it is a shield for the vulnerable. By combining Artificial Intelligence with Government-level verification, we have created a tool that understands not just *what* is being said, but *who* is saying it.

$$Final\ Score = (Legal_{40} + AI_{30} + Infra_{30}) \rightarrow \text{🛡️}\ Trust\ Verified$$

***** End of Documentation *****